FIITJEE

# ALL INDIA TEST SERIES

## CONCEPT RECAPITULATION TEST – IV

## JEE (Main)-2023

### TEST DATE: 27-03-2023

**Time Allotted: 3 Hours** **Maximum Marks: 300**

**General Instructions:**

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

  **Section-A (01 – 20, 31 – 50, 61 – 80)** *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  ***Section-B (21 – 30, 51 – 60, 81 – 90)*** *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*



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# Physics

## PART – A

### SECTION – A

**(Single Choice Answer Type)**

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. The frequency changes by 10% as the source approaches a stationary observer with constant speed $v_s$. What should be the percentage change in frequency as the source recedes from the observer with the same speed? Given that $v_s << v$:
(A) 14.3% (B) 20%
(C) 16.7% (D) 10%

2. Suppose potential energy between electron and proton at separation $r$ is given by $U = k \ln r$, where $k$ is constant. For such hypothetical hydrogen atom, the ratio of energy difference between energy levels ($n = 1$ and $n = 2$) and ($n = 2$ and $n = 4$) is
(A) 1 (B) 2
(C) $\frac{1}{2}$ (D) 3

3. A piece of wax weighs x g in air. A piece of metal is found to weigh y g in water. It is tied to the wax and both together weigh z g in water. Then, the specific gravity of wax is: (z > y)
(A) $\frac{x}{y}$ (B) $\frac{y}{x}$
(C) $\frac{x}{x-(z-y)}$ (D) $\frac{x}{x-z}$

4. Two bodies of masses $M_1 = m$ and $M_2 = 4m$ are placed at a distance $r$. The gravitational potential at a point on the line joining them where the gravitational field is zero is
(A) zero (B) $-\frac{4Gm}{r}$
(C) $-\frac{6Gm}{r}$ (D) $-\frac{9Gm}{r}$

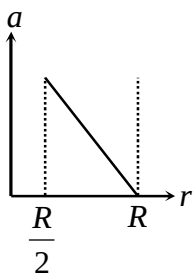
5. The masses and radii of the earth and moon are $M_1$, $R_1$ and $M_2$, $R_2$ respectively. Their centres are at distance $d$ apart. The minimum speed with which a particle of mass $m$ should be projected from a point midway the two centres so as to escape to infinity is
(A) $\sqrt{\frac{2G(M_1+M_2)}{d}}$ (B) $\sqrt{\frac{4G(M_1+M_2)}{d}}$
(C) $\sqrt{\frac{4GM_1M_2}{d}}$ (D) $\sqrt{\frac{4(M_1+M_2)}{d}}$

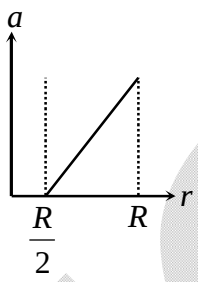
6. A straight rod of length $L$ extends from $x = a$ to $x = L + a$. The gravitational force exerted by it on a point mass $m$ at $x = 0$ if the linear density of rod is $\mu = A + Bx^2$ (where $x$ is $x$ coordinates) is
(A) $Gm\left[\frac{A}{a}+BL\right]$ (B) $Gm\left[A\left(\frac{1}{a}-\frac{1}{a+L}\right)+BL\right]$
(C) $Gm\left[BL+\frac{A}{a+L}\right]$ (D) $Gm\left[BL-\frac{A}{a}\right]$

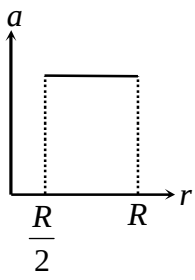


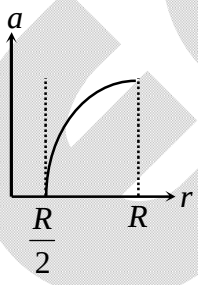
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7. A frictionless tunnel is dug along a chord of the earth at a perpendicular distance $\frac{R}{2}$ from the centre of earth (where *R* is radius of earth). An object is released from one end of the tunnel. The correct graph, showing the variation of acceleration of particle with its distance *r* from centre of earth is

(A)


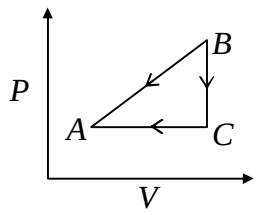

(B)


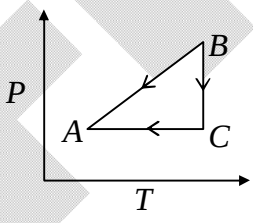

(C)


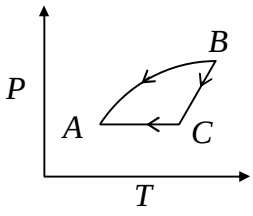

(D)


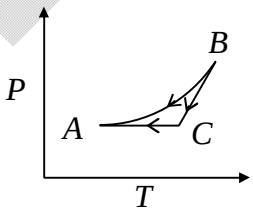

8. Equal amount of same gas in two similar cylinders *A* and *B*, compressed to same final volume from same initial volume one adiabatically and another isothermally, respectively then
(A) final pressure in *A* is more than in *B*
(B) final pressure in *B* is greater than in *A*
(C) final pressure in both able equal
(D) for the gas, value of $\gamma = \frac{C_p}{C_V}$ is required

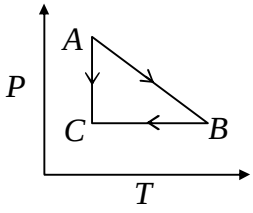
9. *P*-*T* curve of the following process will be




(A)



(B)



(C)


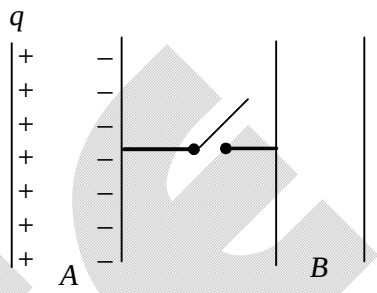

(D)




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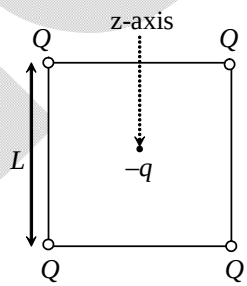
10. On the surface of lake when the atmospheric temperature is –15ºC, 1.5cm thick layer of ice is formed in 20 minutes, time taken to change its thickness from 1.5 cm to 3 cm will be
(A) 20 minutes
(B) less than 20 minutes
(C) greater than 40 minutes
(D) less than 40 minutes and greater than 20 minutes

11. Consider the situation shown in the figure. The capacitor *A* has a charge *q* on it whereas *B* is un charged. The charge appearing on the capacitor *B* a long time after the switch is closed is

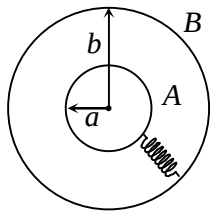



(A) zero (B) $\frac{q}{2}$
(C) $q$ (D) $2q$

12. Four positive charges of same magnitude *Q* are placed at the four corners of a rigid square frame as shown in the figure. The plane of the frame is perpendicular to z-axis. If a negative charge $-q$ is placed at a distance *z* away from the centre above frame ($z << L$) then
(A) negative charge oscillates along the *z*-axis
(B) it moves away from the frame
(C) it moves slowly towards the frame and stay in the plane of the frame
(D) it passes through the frame only once

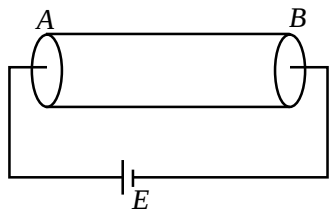



13. Two spherical conductors *A* and *B* of radii *a* and *b* ($b > a$) are placed concentrically in air. The two are connected by a copper wire as shown in figure. Then the equivalent capacitance of the system is




(A) $\frac{4\pi\varepsilon_0 ab}{b-a}$ (B) $4\pi\varepsilon_0(a+b)$
(C) $4\pi\varepsilon_0 b$ (D) $4\pi\varepsilon_0 a$

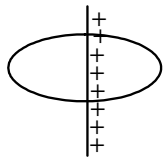
14. A cylindrical conductor *AB* of length *l* and area of cross-section *a* is connected to a battery having emf *E* and negligible internal resistance. The specific conductivity of cylindrical conductor varies as $\sigma = \sigma_0 \frac{l}{\sqrt{x}}$, where $\sigma_0$ is constant and *x* is distance from end *A*.




What is the electric field just near the end *B* of cylinder?
(A) $\frac{E}{l}$ (B) $\frac{3E}{2l}$
(C) $\frac{2E}{3l}$ (D) $\frac{E}{2l}$


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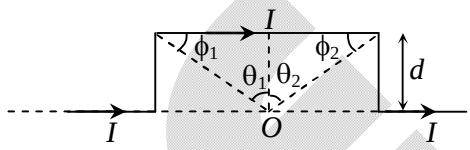
15. A very long uniformly charged rod falls with a constant velocity $V$ through the center of a circular loop. Then the magnitude of induced emf in loop is (charge per unit length of rod = $\lambda$)



(A) $\frac{\mu_0}{2\pi}\lambda V^2$ (B) $\frac{\mu_0}{2}\lambda V^2$

(C) $\frac{\mu_0}{2\lambda}V$ (D) zero

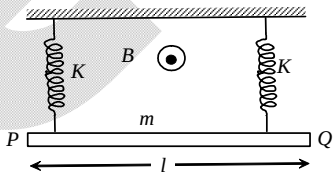
16. The magnitude at $O$ due to current in the infinite wire forming a loop as shown in the following figure is




(A) $\frac{\mu_0 I}{4\pi d}(\cos\phi_1 + \cos\phi_2)$ (B) $\frac{\mu_0}{4\pi}\times\frac{2I}{d}$

(C) $\frac{\mu_0 I}{4\pi d}(\sin\phi_1 + \sin\phi_2)$ (D) $\frac{\mu_0}{4\pi}\times\frac{I}{d}$

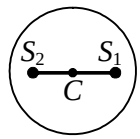
17. A conducting rod of mass $m$ and length $l$ is connected by two identical springs as shown in the figure. Initially the system is in equilibrium. A uniform magnetic field of magnitude $B$ directed perpendicular to the plane of the paper outwards also exists in the region. If a current $I$ is switched on that passes from $P$ to $Q$ through the rod. Further maximum elongation in the spring is [Given: $|mg| = |BIl|$]




(A) $\frac{BIl}{K}$ (B) $\frac{BIl}{4K}$

(C) $\frac{BIl}{8K}$ (D) $\frac{BIl}{16K}$

18. Figure shows two coherent sources $S_1$ and $S_2$ which emits sound of wavelength $\lambda$ in phase. The separation between the sources is $2\lambda$. A circular wire of large radius is placed in such a way that $S_1$ $S_2$ lies in its plane and the middle point of $S_1S_2$ is at the centre of wire. The number of point on the wire where the constructive interference takes place is




(A) 4 (B) 8

(C) 6 (D) 10

19. The $x$-$z$ plane separates two media $A$ and $B$ of refractive indices $\mu_1$ = 1.5 and $\mu_2$ =2. A ray of light travels from $A$ to $B$. Its directions in the two media are given by unit vectors $\vec{u}_1 = a\hat{i} + b\hat{j}$ and $\vec{u}_2 = c\hat{i} + d\hat{j}$. Then

(A) $\frac{a}{c} = \frac{4}{3}$ (B) $\frac{a}{c} = \frac{3}{4}$

(C) $\frac{b}{d} = \frac{4}{3}$ (D) $\frac{b}{d} = \frac{3}{4}$



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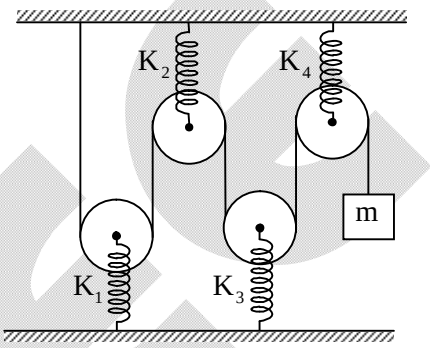
20. A sample contains 16 g of a radioactive material, the half-life of which is 2 days. After 32 days the amount of radioactive material left in the sample is
(A) Less than 1 *mg* (B) (1/4) *g*
(C) (1/2) *g* (D) 1 *g*

## SECTION – B

### (Numerical Answer Type)

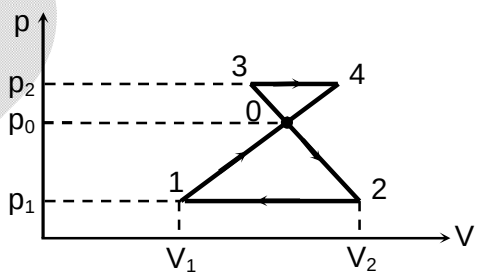
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. In the arrangement as shown in the figure, pulleys are small and springs are ideal. $K_1 = K_2 = K_3 = K_4 = 10$ N/m are force constants of the springs, m = 10 kg. If the time period of small vertical oscillations of the block of mass m is given by $2\pi$ X second, then find the value of X.

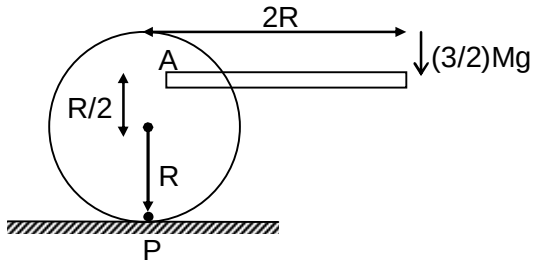



22. A railway tank wagon with its 2m diameter and 6m long horizontal cylindrical body, half full of petrol is driven around a curve of radius 100 m, at a speed of 8.33 m/s. The curve runs smoothly into a straight track and the train maintains a constant speed. If value of nf is 2.3. Find the value of 'n'. neglect viscosity and consider petrol as a solid semi cylinder sliding inside the tank. Given: $\tan^{-1}(0.07) \approx 4°$.

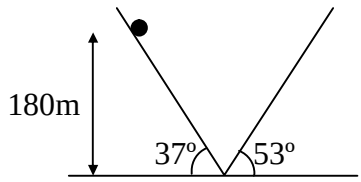
23. Work done by an ideal gas during a closed cycle $1 \to 4 \to 3 \to 2 \to 1$ shown in Figure, if $p_1 = 10^5$ Pa, $p_0 = 3 \times 10^5$ Pa, $p_2 = 4 \times 10^5$ Pa, $V_2 - V_1 = 10$ *l*, and segments 4-3 and 2-1 of the cycle are parallel to the V-axis is $\frac{k \times 10^3}{4}$ J. Find the value of K.




24. A disc of mass M and radius R is placed a rough horizontal surface with its axis horizontal. A light rod of length '2R' is fixed to the disc at point 'A' as shown in figure and a force $\frac{3}{2}Mg$ is applied at the other end. If disc starts to roll without slipping find the value of "$10 \times \mu_{min}$" where $\mu_{min}$ is minimum coefficient of friction between disc and horizontal surface require for pure rolling



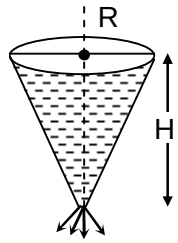

25. Find the time period of the motion of the particle shown in figure.(Neglect the small effect of the bend near the bottom)





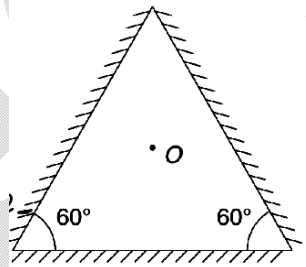

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26. A conical container of radius R = 1m and height H = 5 m is filled completely with liquid. There is a hole at the bottom of container area $\pi \times 10^{-3}$ m$^2$ (see figure). Time taken to empty the conical container is $n \times 200\,\text{sec}$. Then find the value of '$n$'. Take g = 10 m/s$^2$.




27. Two sources of sound are moving in opposite directions with velocities $V_1$ and $V_2$ ($V_1 > V_2$). Both are moving away from a stationary observer. The frequency of both the source is 1700 Hz. What is the value of ($V_1 - V_2$) in m/s so that the beat frequency observed by the observer is 10 Hz. $V_{sound}$ = 340 m/s and assume that $V_1$ and $V_2$ both are very much less than $V_{sound}$.

28. Three plane mirrors are kept as shown in the figure A point object (O) is kept at the centroid of the triangle seen in the figure. Numbers of images formal are




29. A bullet is fired vertically upwards with velocity v from the surface of a spherical planet. When it reaches its maximum height, its acceleration due to the planet's gravity is $\frac{1}{4}$th of its value of the surface of the planet. If the escape velocity from the planet is $v_{esc} = v\sqrt{N}$, then the value of N is (ignore energy loss due to atmosphere)

30. The amount of work done to increase the temperature of one mole of an ideal gas by 30°C is 10x cal. Find the value of x, if it is expanding under the condition $V \propto T^{2/3}$. (R = 2 cal/mol-K)



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# Chemistry

## PART – B

### SECTION – A

**(Single Choice Answer Type)**

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

31. The product 'C' in the following reaction is

$$C_6H_5NO_2 \xrightarrow{Fe/HCl} A \xrightarrow[HCl,0^oC]{NaNO_2^+} B \xrightarrow[\Delta]{H_2O/H^+} C$$

(A) Aniline (B) Benzene
(C) Nitrobenzene (D) Phenol

32. Which of the following ions is paramagnetic?
(A) $La^{+3}$ (B) $Gd^{+3}$
(C) $Lu^{+3}$ (D) $Ce^{+4}$

33. The monomer of glyptal are:
(A) Phenol & formaldehyde (B) Styrene
(C) Ethylene glycol & phthalic acid (D) Ethylene glycol & terephthalic acid

34. $CO(g) + 2H_2(g) \xrightarrow[300^oC]{Catalyst} CH_3OH(g)$

The catalyst used in above reaction is:-
(A) Ni (B) Pt
(C) Cu/ZnO – $Cr_2O_3$ (D) Zn

35. Which of the following is bactericidal antibiotics?
(A) Erythromycin (B) Ofloxacin
(C) Tetracycline (D) Chloramphenicol

36. Which of the following is not a common component of photochemical smog?
(A) Peroxyacetyl nitrate (B) Ozone
(C) Nitric oxide (D) CFC's

37. Which of the following is correct order of acidity?
(A) $CH_3COOH > HCOOH > CNCH_2COOH > CF_3COOH$
(B) $CF_3COOH > CNCH_2COOH > CH_3COOH > HCOOH$
(C) $CNCH_2COOH > CH_3COOH > CF_3COOH > HCOOH$
(D) $CF_3COOH > CNCH_2COOH > HCOOH > CH_3COOH$

38. The molar solubility of $Ni(OH)_2$ in 0.10 M NaOH when the ionic product of $Ni(OH)_2$ is $2.0 \times 10^{-15}$, is:
(A) $2 \times 10^{-15}$ M (B) $2 \times 10^{-14}$M
(C) $2\times 10^{-13}$ M (D) $2 \times 10^{-12}$ M

39. Which of the following statement is correct for emulsion?
(A) They can be diluted by dispersed phase
(B) They do not show Brownian movement
(C) They can be broken into constituents by centrifuging
(D) They do not show tyndall effect



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40. CsBr crystallizes in a BCC lattice. The unit cell length is 436.6 pm. Given that atomic mass of Cs = 133 and that of Br = 80 amu and Avogadro number being $6.023 \times 10^{23}$ $mol^{-1}$, the density of CsBr is:-
(A) 42.5 g $cm^{-3}$ (B) 0.425 $cm^{-3}$
(C) 8.25 g $cm^{-3}$ (D) 4.55 g $cm^{-3}$

41. Which of the following reactions is not involved in thermite process?
(A) $3\ MnO_4 + 8\ Al + \longrightarrow 3\ Mn + 4Al_2O_3$
(B) $Cr_2O_3 + 2l \longrightarrow Al_2O_3 + 2Cr$
(C) $2Fe + Al_2O_3 \longrightarrow 2Al + Fe_2O_3$
(D) $2Al + B_2O_3 \longrightarrow 2B + Al_2O_3$

42. For the non- stoichiometric reaction
$2A + B \rightarrow C + D$
the following kinetic data where observed in three separate experiments, all at 298 K:

| Initial concentration (A) | Initial Concentration (B) | Initial rate of formation (C) |
|---|---|---|
| 0.2 M | 0.2 M | $1.6 \times 10^{-3}$ mol $L^{-1}$ $S^{-1}$ |
| 0.2 M | 0.3 M | $1.6 \times 10^{-3}$ mol $L^{-1}$ $S^{-1}$ |
| 0.4 M | 0.2 M | $3.2 \times 10^{-3}$ mol $L^{-1}$ $S^{-1}$ |

The rate of reaction will be:
(A) $\frac{dC}{dt} = k[A][B]^2$ (B) $\frac{dC}{dt} = k[A]$
(C) $\frac{dC}{dt} = k[A][B]$ (D) $\frac{dC}{dt} = k[A]^2[B]$

43. Given, standard electrode potential;
$Fe^{+3} + 3e^- \longrightarrow Fe,\quad E^o = -0.036\,V$
$Fe^{+2} + 2e^- \longrightarrow Fe,\quad E^o = -0.440\,V$
The standard electrode potential $E^o$ for
$Fe^{+3} + e^- \longrightarrow Fe^{+2}$
(A) –0.476 V (B) –0.404 V
(C) 0.440 V (D) +0.772 V

44. A gas is expanded from $V_1$ to $V_2$ through three different processes:
I. Reversible adiabatic
II. Reversible isothermal
III. Irreversible adiabatic (against a constant external pressure $P_{ext}$)
The correct option is
(A) $[(T_f)_{gas}]_{reversible\ isothermal} > [(T_f)_{gas}]_{Rev\text{-}adiabatic} > [(T_f)_{gas}]_{Irr.adiabatic}$
(B) $[(T_f)_{gas}]_{rev\text{-}isothermal} > [(T_f)_{gas}]_{irr.adiabatic} > [(T_f)_{gas}]_{rev.adiabatic}$
(C) $W_{rev.isothermal} > W_{irrrev.adiabtic} > W_{rev.adiabatic}$
(D) $(P_f)_{rev.isothermal} > (p_f)_{rev.adiabtic} > (P_f)_{irr.adiabatic}$

45. The pressure of $CO_2$ gas at 700 K in the heterogenous equilibrium reaction
$CaCO_3(s) \rightleftharpoons CaO\ (s) + CO_2\ (g)$,
If $\Delta G^o = 130.2$ kJ $mol^{-1}$.
[Given, antilog (0.29)= 1.95]
(A) $1.95 \times 10^{-13}$ atm (B) $1.95 \times 10^{-10}$ atm
(C) $1.95 \times 10^{-11}$ atm (D) $1.95 \times 10^{-9}$ atm


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46. Select incorrect option

(A) $Ca(OH)_2$ $\xrightarrow{CO_2}$ white turbidity; $\xrightarrow{CO_3^{-2}}$ white ppt.

(B) $MnO_4^-/H^+ \xrightarrow{SO_2}$ purple colour decolourises

(C) $Pb(NO_3)_2 \xrightarrow{H_2S}$ Black ppt.

(D) $Na_2[Fe(CN)_5NO]_{aq.} \xrightarrow{H_2S}$ red solution

47. The order of basic strength of given oxide:

(A) $Na_2O > MgO > Al_2O_3 > CuO$
(B) $MgO > Al_2O_3 > CuO > Na_2O$
(C) $Al_2O_3 > MgO > CuO > Na_2O$
(D) $CuO > Na_2O > MgO > Al_2O_3$

48. The incorrect statement is:

(A) $BF_4^-$ has central atom $sp^3$ hybridised
(B) $SiO_2$ has $sp^2$ hybridised central atom
(C) $BH_3$ is a Lewis acid
(D) Silicone is a polymer

49. In the following redox equation, number of electrons exchanged between redox half cells after balancing are:

$$[Cr(OH)_4]^{\ominus} + H_2O_2 \xrightarrow[\text{medium}]{\text{basic}} CrO_4^{-2} + H_2O$$

(A) 5
(B) 4
(C) 6
(D) 3

50. The strongest reducing agent among the hydrides of group 15 elements is

(A) $NH_3$
(B) $AsH_3$
(C) $BiH_3$
(D) $PH_3$

## SECTION – B

**(Numerical Answer Type)**

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. No. of statements that are true in the following are:

(A) Halide ligands forms high spin complex
(B) Strong ligands form low spin complex
(C) $[FeF_6]^{-3}$ is inner orbital complex
(D) $[NiCl_4]^{-2}$ is outer orbital complex

52. Number of radial nodes for 4d-orbital are_____.

53. From the following properties, number of properties of alkali metals increases as atomic number increases:

(a) Metallic character
(b) Ionic radius
(c) Melting point
(d) Density
(e) Ionization enthalpy



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54. Match List-I and List-II

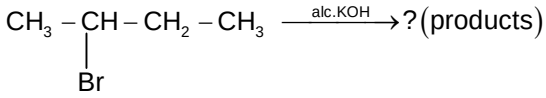
| List-I (Reagent) | | List-II (Used as reagent for) |
|---|---|---|
| (A) | Ammonical $AgNO_3$ | (i) primary amine |
| (B) | $HIO_4$ | (ii) Aldehyde |
| (C) | Alkaline $KMnO_4$ | (iii) Vicinal –OH group |
| (D) | Chloroform + NaOH | (iv) Double bond |

Which of the following is incorrectly matched:

(1) A $\rightarrow$ ii (2) C $\rightarrow$ iv

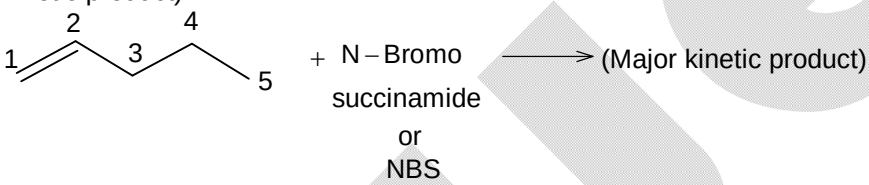
(3) D $\rightarrow$ i (4) A $\rightarrow$ i

55. Number of products (including stereoisomers) for the following $E_2$ (elimination bimolecular) reaction, are:




56. If $Hg_2Cl_2$ is 50% dissociated, then find out its van't Hoff factor

57. Identify the position at which the following reaction will take place & 'Br' will be substituted (major kinetic product):-




58. 100 g sample of '147%' oleum was taken and calculated amount of $H_2O$ was added to make $H_2SO_4$. 500 mL solution of x M KOH solution is required to neutralise the solution. The value of x is_____.

59. Calculate the (in atm) pressure exerted by one mole of $CO_2$ gas at 273 K in a container of volume 22.42 L, if the van der Waal constant a = 3.592 $L^3$ atm $mol^{-2}$. Assume that the volume occupied by $CO_2$
Molecules are negligible.

60. Which of the following compounds have peroxy linkage.
$H_3PO_5$, $H_2SO_5$, $H_2S_2O_8$, $H_4P_2O_8$, $HNO_3$, $H_2S_2O_3$, $H_2S_2O_6$



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# Mathematics

## PART – C

### SECTION – A

**(Single Choice Answer Type)**

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

61. If $\lim\limits_{x\to 0}\int\limits_0^x \frac{t^2 dt}{(x-\sin x)\sqrt{a+t}} = 1$, then the value of a is

(A) 4 (B) 2
(C) 1 (D) none of these

62. The length of the common chord of the ellipse $\frac{(x-1)^2}{9}+\frac{(y-2)^2}{4}=1$ and the circle $(x-1)^2 + (y-2)^2 = 1$.

(A) 2 (B) $\sqrt{3}$
(C) 4 (D) None of these

63. If all the real solutions of the equation $4^x - (a-3)2^x + (a-4) = 0$ are non positive, then

(A) $4 < a \le 5$ (B) $0 < a < 4$
(C) $a > 4$ (D) $a < 3$

64. If $a\sin x + b\cos(C+x) + b\cos(C-x) = \alpha$, then the minimum value of $|\cos C|$ is

(A) $\sqrt{\frac{\alpha^2-a^2}{b^2}}$ (B) $\sqrt{\frac{\alpha^2-a^2}{4b^2}}$
(C) $\sqrt{\frac{\alpha^2-a^2}{16b^2}}$ (D) none of these

65. If $x_i > 0$, i = 1, 2, …., 50 and $x_1 + x_2 + \ldots + x_{50} = 50$, then the minimum value of $\frac{1}{x_1}+\frac{1}{x_2}+\ldots+\frac{1}{x_{50}}$ equals to

(A) 50 (B) $(50)^2$
(C) $(50)^3$ (D) $(50)^4$

66. The coefficient of $x^n$ in $\left(1-x+\frac{x^2}{2!}-\frac{x^3}{3!}+\ldots.+\frac{(-1)^n x^n}{n!}\right)^2$ is

(A) $\frac{(-n)^n}{n!}$ (B) $\frac{(-2)^n}{n!}$
(C) $\frac{1}{(n!)^2}$ (D) $-\frac{1}{(n!)^2}$

67. If $|z| < 4$, then $|iz + 3 - 4i|$ is less than

(A) 4 (B) 5
(C) 6 (D) 9



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68. The number of permutations of the letters of the word HINDUSTAN such that neither the pattern 'HIN' nor 'DUS' nor 'TAN' appears, are
(A) 166674 (B) 169194
(C) 166680 (D) 181434

69. A and B are two events such that P(A) = 0.2 and P(A∪B) = 0.7. If A and B are independent events then P(B′) equals
(A) 2/7 (B) 7/9
(C) 3/8 (D) none of these

70. If $f(\theta) = \frac{(2\cos\theta - 1)(2\cos 2\theta - 1)(2\cos 4\theta - 1)\cdots(2\cos 2^{n-1}\theta - 1)}{2\cos 2^n\theta + 1}$ for $n \in N$ and $\theta \neq 2m\pi \pm \frac{2\pi}{3}$, $m \in I$, then $f(\pi/4)$ is equal to
(A) $\sqrt{2} - 1$ (B) $1 - \sqrt{2}$
(C) $\sqrt{3} - 1$ (D) $1 - \sqrt{3}$

71. Solution set of $[\sin^{-1}x] > [\cos^{-1}x]$, where [.] denotes greatest integer function
(A) [sin1, 1] (B) $\left[\frac{1}{\sqrt{2}}, 1\right]$
(C) (cos1, sin1) (D) None of these

72. Let f(x) be continuous at x = 0 and f″ (0) = 4, the value of $\lim_{x\to 0} \frac{2f(x) - 3f(2x) + f\,4x}{x^2}$ is
(A) 11 (B) 2
(C) 12 (D) None of these

73. Let $f_p(\alpha) = \left(\cos\frac{\alpha}{p^2} + i\sin\frac{\alpha}{p^2}\right)\left(\cos\frac{2\alpha}{p^2} + i\sin\frac{2\alpha}{p^2}\right) \ldots \left(\cos\frac{\alpha}{p} + i\sin\frac{\alpha}{p}\right)$. Then $\lim_{n\to\infty} [\,|\, f_n(2\pi) \,|\,]$ is equal to, ([.] denotes the greatest integer function)
(A) –1 (B) 1
(C) i (D) –i

74. The value of k for which two tangents can be drawn from (k, k) to the circle $x^2 + y^2 + 2x + 2y - 16 = 0$ is
(A) $k \in R^+$ (B) $k \in R^-$
(C) $k \in (-\infty, -4) \cup (2, \infty)$ (D) $k \in (0, 1]$

75. On the interval $\left[\frac{5\pi}{4}, \frac{4\pi}{3}\right]$, the least value of the function $f(x) = \int_{5\pi/4}^{x} (3\sin t + 4\cos t)dt$ is
(A) $\frac{3}{2} + \frac{1}{\sqrt{2}} - 2\sqrt{3}$ (B) $\frac{3}{2} - \frac{1}{\sqrt{2}} + 2\sqrt{3}$
(C) $\frac{3}{2} - \frac{1}{\sqrt{2}} - 2\sqrt{3}$ (D) none of these


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76. If $f(x) = \int_x^{x^2} (t-1)\,dt$, $1 \le x \le 2$, then global maximum value of f(x) is

(A) 1 (B) 2
(C) 3 (D) 4

77. $\int_0^{\infty} \left[2e^{-x}\right]dx$ (where [ ] denotes the greatest integer function) is equal to

(A) 0 (B) In2
(C) $e^2$ (D) $2e^{-1}$

78. If $I_1 = \int_0^{\pi/2} \frac{\cos^2 x}{1+\cos^2 x}dx$, $I_2 = \int_0^{\pi/2} \frac{\sin^2 x}{1+\sin^2 x}dx$, $I_3 = \int_0^{\pi/2} \frac{1+2\cos^2 x \sin^2 x}{4+2\cos^2 x \sin^2 x}dx$, then

(A) $I_1 = I_2 > I_3$ (B) $I_3 > I_1 = I_2$
(C) $I_1 = I_2 = I_3$ (D) none of these

79. The curve, which satisfies the differential equation $\frac{xdy - ydx}{xdy + ydx} = y^2 \sin(xy)$ and passes through (0, 1), is given by

(A) y (1 - cos xy) + x = 0 (B) sinxy - x = 0
(C) siny + y = 0 (D) cosxy - 2y = 0

80. If $\Delta(x) = \begin{vmatrix} x & 1+x^2 & x^3 \\ \log(1+x^2) & e^x & \sin x \\ \cos x & \tan x & \sin^2 x \end{vmatrix}$ then

(A) $\Delta(x)$ is divisible by x (B) $\Delta(x) = 0$
(C) $\Delta'(x) = 0$ (D) None of these

## SECTION – B

**(Numerical Answer Type)**

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. If the circle $x^2 + y^2 + 4x + 22y + l = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x + 8y - m = 0$, then l + m is equal to

82. If line $y = 2x + \frac{1}{4}$ is tangent to $y^2 = 4ax$, then 2a is equal to

83. The minimum value of $3\tan^2\theta + 12\cot^2\theta$ is

84. The value of $\sin^{-1}\left[\cot\left(\sin^{-1}\sqrt{\frac{2-\sqrt{3}}{4}} + \cos^{-1}\frac{\sqrt{12}}{4} + \sec^{-1}\sqrt{2}\right)\right]$ is equal to


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85. If S denote the sum to infinity and $S_n$ the sum of n terms of the series $1+\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+\ldots$ such that $S - S_n < \frac{1}{300}$, then the least value of n is

86. The area enclosed between the curves $y = \sin^2 x$ and $y = \cos^2 x$ in the interval $0 \le x \le \pi$ is A, find the value of 2A.

87. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = x_1\hat{i} + x_2\hat{j} + x_3\hat{k}$, where $x_1, x_2, x_3 \in \{-3, -2, -1, 0, 1, 2\}$.
Number of possible vectors $\vec{b}$ such that $\vec{a}$ and $\vec{b}$ are mutually perpendicular, is

88. Shortest distance between lines $\frac{x-6}{1} = \frac{y-2}{-2} = \frac{z-2}{2}$ and $\frac{x+4}{3} = \frac{y}{-2} = \frac{z+1}{-2}$ is

89. Find the length of the perpendicular from the point (1, 2, 3) to the line $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$.

90. If A + B + C = $\pi$, then the value of determinant
$\begin{vmatrix} \sin^2 A & \cot A & 1 \\ \sin^2 B & \cot B & 1 \\ \sin^2 C & \cot C & 1 \end{vmatrix}$ is equal to


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